SPATIAL TAXONOMY · FLOOR PLATE METHODOLOGY

Key Plans and Tiles

A geometric self-similar aperiodic space-planning system. Furniture-driven Key Plans aggregate into Tiles; Tiles compose Floor Plates without remainder. Methodology V12.

AGENDA — SEVEN SECTIONS

SECTION 01

What “Key Plans and Tiles” Means

WORKING DEFINITION, FROM THE METHODOLOGY MEMO

“**Key Plans and Tiles**” means a geometric self-similar aperiodic space-planning system based on furniture/equipment arrangements and circulation versus modular area-per-person progressions. Various Key Plans are summed up together in order to make Tiles. Not all the Key Plans are supposed to fit in all the Tiles — in the case of the building corners or areas surrounding the Elevator Lobbies, there may be “unique” Special Tiles where either only certain Key Plans fit or the Key Plans found within these Special Tiles are exclusive to these areas. However, in all cases, the entirety of the Floor Plates is dividable by a certain combination of Key Plans, with no remainder of “free space” that is not represented by a Key Plan.

Translate: a Key Plan is the smallest unit — a labelled rectangle around one tenant's actual furniture, sized to enclose desks, storage, circulation, and accessibility. A Tile is a composable group of Key Plans. The Floor Plate is the assembly of Tiles. Nothing on the floor is unrepresented.

The system is ***aperiodic*** because Tiles do not repeat on a fixed grid. It is ***self-similar*** because each Tile is exactly one HVAC climate zone, regardless of size. It is ***geometric*** because all dimensions resolve to real metric measurements driven by real furniture SKUs.

SECTION 02

The Tile Catalogue — Small Tiles

EIGHT TILE TYPES · 2,700 SF EACH

Small Tile Family — 2,700 SF (250.84 m²) each. Each Tile is one climate zone; combinations aggregate climate-control authority. Source: `tile-system.dtcg.json`.

TILE-A

Tile A — Corporate Office

2,700 SF / 250.84 m²

Single corporate tenant unit (2,700 SF). Maximum climate-zone autonomy.

1 climate zone

TILE-B1

END CAP

Tile B-1 — Private Office End Cap

2,700 SF / 250.84 m²

Five private offices (300 + 450 + 300 + 450 + 300 SF) plus 900 SF corridor.

5 climate zones

TILE-C1

Tile C-1 — Professional (Med + Small)

2,700 SF / 250.84 m²

Professional Office Medium (2,000 SF) + Small (700 SF).

2 climate zones

TILE-C2

Tile C-2 — Professional (Med Pair)

2,700 SF / 250.84 m²

Two Medium Professional Office units (1,100 SF each) + 500 SF corridor.

2 climate zones

TILE-C3

Tile C-3 — Professional (Large+Small)

2,700 SF / 250.84 m²

Large Professional (2,400 SF) + Small Professional (300 SF). Lab/Medical anchor.

2 climate zones

TILE-C4

Tile C-4 — Professional Single Large

2,700 SF / 250.84 m²

Single 2,200 SF Professional unit + 500 SF corridor/support. Civic/Academic.

1 climate zone

TILE-E1

END CAP

Tile E-1 — Mixed End Cap (Left)

2,700 SF / 250.84 m²

2 × Private (300 SF) + Professional Small (800 SF) + 1,300 SF corridor.

3 climate zones

TILE-E2

END CAP

Tile E-2 — Mixed End Cap (Right)

2,700 SF / 250.84 m²

Mirror of E-1 for right building end. Handles corner column grid.

3 climate zones

SECTION 03

THREE TILE TYPES · 4,900 SF EACH

The Tile Catalogue — Large Tiles

Large Tile Family — 4,900 SF (455.22 m²) each. Approaches the 1/4 floor tile in scale; useful for corporate anchors, large professional offices, and dense private-office mixes. Source: `tile-system.dtcg.json`.

TILE-F

Tile F — Large Corporate Office

4,900 SF / 455.22 m²

Single 4,900 SF corporate tenant. Approaches 1/4 floor tile.

1 climate zone

TILE-G

Tile G — Large Private Office Mix

4,900 SF / 455.22 m²

Ten private offices (300–500 SF each) + 900 SF corridor. Maximum climate-zone count.

10 climate zones

TILE-H

Tile H — Large Professional Office

4,900 SF / 455.22 m²

Three Professional Office units (any sub-type mix) + 700 SF corridor.

3 climate zones

SECTION 04

FIVE FRACTIONS OF 20,000 SF

The Size Hierarchy

Five fractions of the 20,000 SF net leasable floor plate. Source: `floor-plate-standards.dtcg.json`.

1/8 Floor	13%	Minimum Corporate Office tile; Private Office aggregate	**2,500 SF**	232.26 m²
1/4 Floor	25%	Standard Professional Office leasehold; mid-tier Corporate	**5,000 SF**	464.52 m²
1/2 Floor	50%	Tenant Lounge (2nd floor); large Corporate Office option	**10,000 SF**	929.03 m²
3/4 Floor	75%	Large Corporate Office option (non-standard tile composition)	**15,000 SF**	1393.55 m²
Full Floor	100%	Full floor; one Corporate tenant occupies entire upper floor	**20,000 SF**	1858.06 m²

A Corporate Office tenant may occupy any fraction up to the full floor. The 1/8 floor (2,500 SF) is the minimum Corporate Office tile ***and*** the aggregate tile for Private Office leaseholds. The 1/2 floor is dedicated on the second floor to the Tenant Lounge per the design memo.

SECTION 05

20,000 SF · EVERY SF A KEY PLAN

One Floor — Filled Exactly

A worked composition — one example of how tiles fill the floor exactly. Each segment is one Tile, drawn to scale; the composition sums to 20,000 SF with every square foot represented by a Key Plan.

FLOOR PLATE COMPOSITION — 20,000 SF NET LEASABLE · DRAWN TO SCALE

Tile F — 4,900 SF · 1 climate zone

Tile A — 2,700 SF · 1 climate zone

Tile H — 4,900 SF · 3 climate zones

Tile C-1 — 2,700 SF · 2 climate zones

Tile B-1 — 2,700 SF · 5 climate zones

Common — 2,100 SF

Other combinations are valid as long as they sum to 20,000 SF (or 18,850 SF after corridors + restrooms). The methodology requires that whatever combination is chosen, ***every square foot is represented by a Key Plan***.

SECTION 06

FOUR RULES · BIM.TILE.DESIGN-PRINCIPLES

Design Principles

Encoded as token metadata at `bim.tile.design-principles`. Four principles govern every composition.

CLIMATE-ZONE AUTONOMY

One tile, one HVAC zone. A tenant who wants direct thermostat control must lease a whole Tile. Tenants combining tiles must combine climate control with the same negotiated authority. Smaller tiles = more zones = more autonomy per tenant; larger tiles = fewer zones = more contiguous tenant space.

ROLLING EFFICIENCY

The Tile composition must reach 100% floor-plate utilisation at any tenant mix — no free space without a corresponding Key Plan. As tenants come and go, the floor recomposes; the engineering rule is that the recomposition is always achievable from the published catalogue.

CORNER HANDLING

Building short sides and corners use End Cap tiles (B-1, E-1, E-2). Small tiles at corners may create structural-grid boxing issues. Column grid is reviewed before specifying corner tile types; core position is adjusted if short-side tiles conflict.

PROFESSIONAL → CORPORATE GAP

The jump from Professional Office (~5,000 SF / 1/4 tile) to Corporate Office (min 2,500 SF / 1/8 tile) leaves a gap for tenants needing 5,000–7,500 SF. Tile H (4,900 SF) and Small+Small Professional combinations (~4,200 SF) bridge it.

SECTION 07

Leasing Economics & Rolling Efficiency

WHY SMALLER TILES MATTER · HOW THE FLOOR RECOMPOSES

WHY SMALLER TILES MATTER

Climate-zone autonomy as a leasing feature

A small tenant who wants to set their own thermostat must lease a full Tile. The Tile catalogue is therefore not just a geometric system — it is a leasing instrument. A Private Office in Tile B-1 shares climate with four other private offices; a Private Office in Tile G shares with nine; a Corporate Office in Tile F sets its own. The leasing agreement makes this explicit and prices accordingly.

Smaller tiles increase tenant autonomy and operational complexity in parallel. A Private-Office-heavy floor (Composition B in the Building Width Calculator companion deck) can host 40+ independent climate zones on a single 20,000 SF plate. The HVAC engineering is more expensive; the leasing yield is higher because each thermostat is its own line item.

ROLLING EFFICIENCY

Engineering efficiency vs. economic efficiency

The engineering definition of “efficiency” is net-leasable divided by gross-buildable. The Woodfine methodology calls this metric outdated. The replacement metric is **economic efficiency**: the share of a floor that is actively leased and represented by a Key Plan that matches a real tenant's furniture inventory.

A floor at 100% engineering efficiency with one tenant in a 60-foot rectangle, putting bookshelves on top of desks because the Habitat zone is too shallow, is failing the economic test. A floor at 88% engineering efficiency with seven tenants in mixed Tile compositions, each at 100% tenant-utilisation inside their leasehold, is succeeding.

Rolling efficiency is the dynamic version: as tenants change, the Tile recomposition must always be valid. The catalogue is sized to make that always possible.